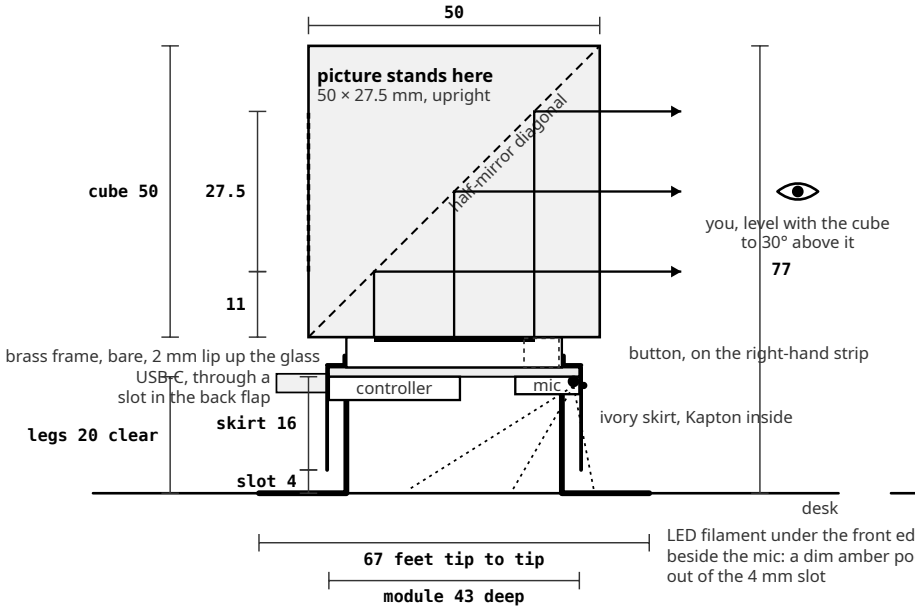
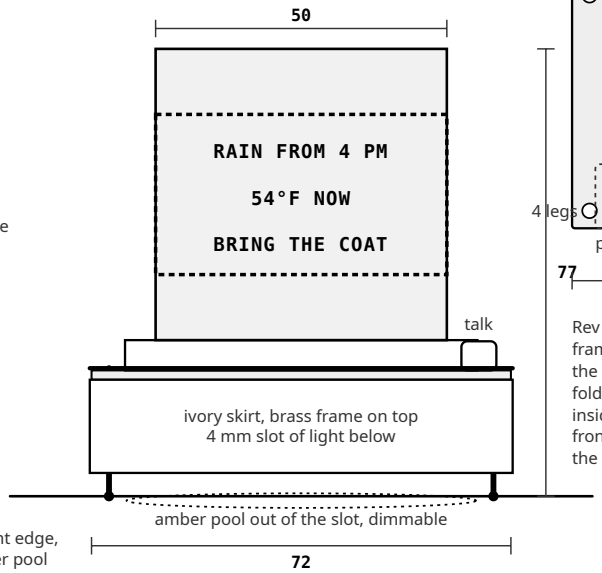
Desk hologram: how the finished thing looks

A 50 mm glass cube on the 2.42" display in a brass-topped ivory skin, four stainless legs, the answer floating inside the glass. 72 × 67 × 77 mm, about 354 g. One scale, 4 px per mm; dimensions in mm. Rev B, 2026-11-07.

Side section, viewer on the right

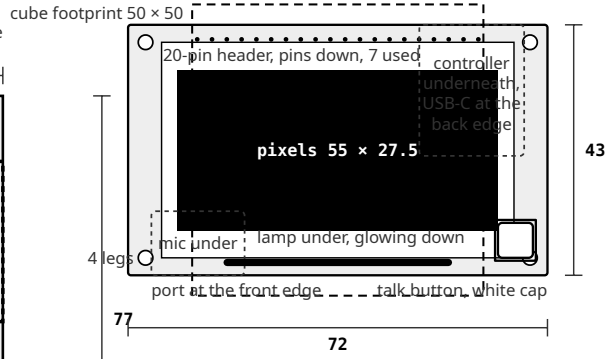


Front view: what you see



Top view, brass frame on top

module see-through; front of the desk at the bottom



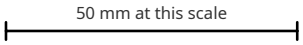
Rev B: the brass sheet lies on the board as a bare frame round the glass; its 2 mm lip turns up against the glass edge, so no green shows. The four flaps fold down 16 mm at the edges, ivory outside, Kapton inside. Holes: four legs, the button (7 × 7) at the front right, the mic port (Ø3) in the front flap, the USB-C (10 × 5) in the back flap.

Key numbers

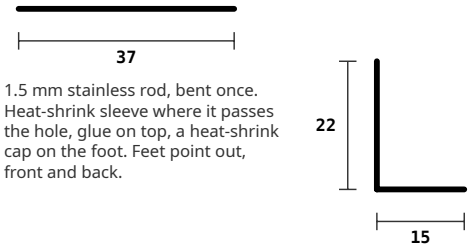
72 × 67 × 77 mm	overall, feet tip to tip
50 mm, 314 g	cube, 50:50, K9 glass
50 × 27.5 mm	picture, 116 × 64 px
3 × 14 or 8 × 19	characters a page, big or small font
4 rods, 0 solder joints	frame: legs through the module's own holes
17 wires, 2 capacitors, 1 resistor	all 30 AWG, under one PCB
3 V filament at 12 mA	amber glow under the front edge, PWM-dimmed
1 sheet of 0.13 mm brass	the skin: bare brass top, ivory skirt 16, slot 4
brass · ivory · grey · brown	the palette: frame, skirt, steel legs, the pool
5 V USB-C, under 0.4 A	power, the owned 2 A adapter
ws :8790	to the desktop's voice server over Wi-Fi

Reading the section

Every pixel sends light straight up. Half of it bounces off the diagonal and leaves through the front face, toward you.
Your eye traces it back: the picture stands at the cube's back face, the display's front edge at its top, 50 × 27.5 mm.
The other half leaks out of the top: from above you see the bare, mirrored text. From the front you see the hologram.
A black card behind the cube (optional) blacks out the room.



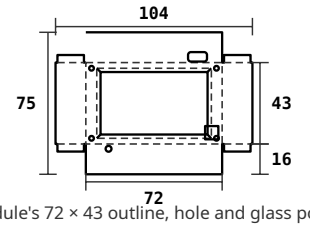
One leg, before and after the bend

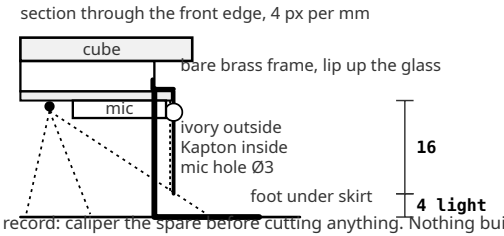


1.5 mm stainless rod, bent once. Heat-shrink sleeve where it passes the hole, glue on top, a heat-shrink cap on the foot. Feet point out, front and back.

The skin, Rev B 2026-11-07: one piece of 0.13 mm brass sheet, folded as a tray

Flat pattern 104 × 75 mm, laid along a K&S 4 × 10 in sheet. Cut the outline and the window, fold the four flaps down 16 mm, the window's 2 mm lip up, the four tabs behind the front and back flaps; solder the tabs. Scuff, wipe, mask the top at the fold, two coats of canvas white on the flaps.
Kapton on every inside face. Top: bare brass round the black glass. Edges: ivory. Below: 4 mm of light.
Card pattern first (1:1 on the build page), brass second. The skin carries nothing; the legs still do.
Right: the flat pattern (solid = cut, dashed = fold) and a section through the front edge.





Rev B, 2026-11-07: the skin added at the builder's word (brass, ivory, grey, brown). The module's 72 × 43 outline, hole and glass positions are the Cyber Dawn record: caliper the spare before cutting anything. Nothing built.